

Convexity, Constraints and Optimality Conditions

Convexity tests

- **Convex set** C : $(1-t)x + ty \in C$ for all $x, y \in C$ and $t \in [0, 1]$. **Convex function** (the Jensen inequality): $f((1-t)x + ty) \leq (1-t)f(x) + tf(y)$.
- **First order** ($f \in C^1$): f is convex $\iff f(y) \geq f(x) + \nabla f(x)^\top (y - x)$ for all x, y ; the tangent lies below the graph. For $f(x) = x^2$: $f(3) = 9 \geq f(1) + f'(1)(3-1) = 5$.
- **Second order** ($f \in C^2$): f is convex $\iff \nabla^2 f(x) \succeq 0$ at every x ; $\nabla^2 f \succ 0$ everywhere implies strict convexity (not conversely). $H \succeq 0$: all eigenvalues ≥ 0 , or **all** principal minors ≥ 0 ; $H \succ 0$: leading minors > 0 . A1: $f = x^2 + axy + 4y^2 + 2x$ has $H_{11} = 2$, $H_{22} = 8$, $\det H = 16 - a^2$, so f is convex $\iff |a| \leq 4$.
- **One violating pair disproves** convexity; passing random tests proves nothing. A2: $f = x^2 y^2$, $p = (1, 0)$, $q = (0, 1)$: $f(p) = f(q) = 0$, but $f(\frac{p+q}{2}) = \frac{1}{16} > 0$.

What preserves convexity

- **Intersections** $\bigcap_\alpha C_\alpha$ of convex sets are convex; **unions** in general are not (two disjoint disks).
- **Sublevel set** $\{x : f(x) \leq \alpha\}$ of a convex f is convex; for a nonconvex f check the set itself: $S = \{(x, y) : x^2 y^2 \leq 1\}$ is not convex. **Epigraph** $\text{epi } f = \{(x, s) : f(x) \leq s\}$ is convex $\iff f$ is convex.
- **Affine maps**: the image $\{Ax + b : x \in C\}$ and the preimage $\{x : Ax + b \in C\}$ of a convex set are convex. Convex: half-spaces, balls, boxes, $\Delta_d = \{x \in \mathbb{R}^d : x \geq 0, \mathbf{1}^\top x = 1\}$.
- The feasible set $\{g_i \leq 0, h_j = 0\}$ with convex g_i and affine h_j is convex. A3: $T = \{x^2 + y^2 \leq 4\} \cap \{y \geq x^2 - 1\}$ is a disk intersected with an epigraph, hence convex.

Projections

- **Projection** $P_C(x) = \arg \min_{y \in C} \|y - x\|$; for a closed convex C it exists and is unique.
- **Ball** $\{\|y - c\| \leq r\}$: $P(x) = c + r(x - c) / \max(r, \|x - c\|)$. **Box** $[l, u]$: $P(x) = \min(\max(x, l), u)$ coordinatewise. B2: $x = (0.9, 1.2)$ goes to $(0.6, 0.8)$ on the unit ball and to $(0.9, 1)$ on the box $[-1, 1]^2$.
- **Probability simplex** Δ_d : sort v in decreasing order into u ; $s_j = u_1 + \dots + u_j$;

$\rho = \max\{j : u_j - (s_j - 1)/j > 0\}$;
 $\theta = (s_\rho - 1)/\rho$; $P(v) = \max(v - \theta, 0)$ in the **original** order. B3: $v = (0.1, 1.3, 0.5)$, $u = (1.3, 0.5, 0.1)$, $s = (1.3, 1.8, 1.9)$, $\rho = 2$, $\theta = 0.4$, $P(v) = (0, 0.9, 0.1)$.

- **Obtuse angle**: $p = P_C(x) \iff p \in C$ and $(x - p)^\top (y - p) \leq 0$ for all $y \in C$, i.e. $x - p$ lies in the normal cone $N_C(p)$. On Δ_d it suffices to test the vertices $y = e_k$.

KKT conditions

- Problem: $\min f(x)$ subject to $g_i(x) \leq 0$, $h_j(x) = 0$. Lagrangian: $\mathcal{L}(x, \lambda, \nu) = f(x) + \sum_i \lambda_i g_i(x) + \sum_j \nu_j h_j(x)$ with $\lambda \geq 0$ and ν of any sign.
- **Karush–Kuhn–Tucker (KKT) conditions** at x^* . *Stationarity*: $\nabla_x \mathcal{L} = \nabla f + \sum_i \lambda_i \nabla g_i + \sum_j \nu_j \nabla h_j = 0$. *Primal feasibility*: $g_i(x^*) \leq 0$, $h_j(x^*) = 0$. *Dual feasibility*: $\lambda_i \geq 0$. *Complementary slackness*: $\lambda_i g_i(x^*) = 0$.
- **Signs**: write every inequality as $g \leq 0$ and add $+\lambda g$ to $\mathcal{L}$ with $\lambda \geq 0$; $x_1 \geq 0$ becomes $-x_1 \leq 0$. SciPy writes inequalities as $c(x) \geq 0$: pass $c = -g$.
- **Geometry**:

$$-\nabla f(x^*) = \sum_{i \text{ active}} \lambda_i \nabla g_i + \sum_j \nu_j \nabla h_j,$$

the antigradient lies in the cone of the active gradients, which under Slater's condition is the normal cone $N_C(x^*) = \{s : s^\top (y - x^*) \leq 0 \forall y \in C\}$. C1: $\min(x_1 - 2)^2 + (x_2 - 1)^2$ subject to $x_1 + x_2 - 1 \leq 0$ gives $x^* = (1, 0)$, $\lambda^* = 2$ and $-\nabla f(x^*) = (2, 2) = 2 \cdot (1, 1)$.

Active constraints

- A constraint is **active** at x^* if $g_i(x^*) = 0$ and **inactive** if $g_i(x^*) < 0$; equalities are always active.
- Complementary slackness: inactive $\Rightarrow \lambda_i = 0$; $\lambda_i > 0 \Rightarrow$ active. An active constraint **may** have $\lambda_i = 0$.
- **Active-set search**: guess the active set; solve stationarity with $g_i = 0$ for active i and $\lambda_i = 0$ for the rest; keep the candidate only if $g(x) \leq 0$ and $\lambda \geq 0$. A negative λ_i means a wrong active set.
- C2: $\min(x_1 - 2)^2 + (x_2 - 1)^2$ subject to $x_1 + x_2 - 4 \leq 0$. The case $\lambda = 0$ gives $x^* = (2, 1)$ with $g = -1 < 0$, so $\lambda^* = 0$. Assuming $g = 0$ gives $\lambda = -1 < 0$: rejected.
- A multiplier is a price: relaxing $g_i \leq 0$ to $g_i \leq u_i$ changes the optimal value at the rate $\partial p^* / \partial u_i = -\lambda_i^*$.

Slater's condition

- **Slater's condition**: the problem is convex (f, g_i convex, h_j affine) and has a strictly feasible point: $g_i(\bar{x}) < 0$, $h_j(\bar{x}) = 0$. C1: $\bar{x} = (0, 0)$, $g(\bar{x}) = -1 < 0$.
- In a convex problem every KKT point is a global minimum; with a Slater point the converse holds too: every minimum has multipliers.
- **Without a constraint qualification KKT may fail**: $\min x$ subject to $x^2 \leq 0$. Only $x^* = 0$ is feasible, stationarity $1 + 2\lambda x^* = 1 \neq 0$ has no solution, and there is no Slater point. Here $q(\lambda) = -\frac{1}{4\lambda} \rightarrow p^*$ as $\lambda \rightarrow \infty$, but no λ attains it.

Duality

- **Dual function** $q(\lambda, \nu) = \inf_x \mathcal{L}(x, \lambda, \nu)$ is concave for every problem (an infimum of affine functions). Dual problem: $d^* = \max_{\lambda \geq 0, \nu} q(\lambda, \nu)$.
- **Weak duality**: $q(\lambda, \nu) \leq f(x)$ for feasible x and $\lambda \geq 0$, so $d^* \leq p^*$ and the duality gap $p^* - d^* \geq 0$. C1: $q(\lambda) = 2\lambda - \frac{\lambda^2}{2}$ and $q(1) = 1.5 \leq p^* = 2$.
- **Strong duality**: convex + Slater $\Rightarrow p^* = d^*$. C1: $\max q = q(2) = 2 = p^*$, attained at the KKT multiplier λ^* .
- **Gap without convexity**: $\min x_1 x_2 + x_2 x_3 + x_1 x_3$ subject to $x_i^2 = 1$ has $p^* = -1$ and $d^* = -\frac{3}{2}$, a gap of $\frac{1}{2} > 0$.
- **Caveat (S-lemma)**: a problem with a quadratic objective, one quadratic constraint and a strictly feasible point has a zero gap even when nonconvex.

Top traps

- Classify eigenvalues μ_k with a **relative** tolerance $\varepsilon \max_k |\mu_k|$: an absolute one turns a tiny $H \succ 0$ into zero.
- A cross term $c x_i x_j$ puts c into **both** H_{ij} and H_{ji} ; $c/2$ is the entry of A in $x^\top A x$.
- Leading minors decide only $H \succ 0$; $H \succeq 0$ needs **all** principal minors: $\text{diag}(0, -1)$ has zero leading minors but is not $\succeq 0$.
- Simplex: the sort only finds θ ; return $\max(v - \theta, 0)$ in the **original** order. Every coordinate $\leq \theta$ vanishes, positive ones too: in B3 $v_1 = 0.1 > 0$ but $x_1 = 0$.
- Signs: $g \leq 0, +\lambda g, \lambda \geq 0$ (SciPy: $c(x) \geq 0$); ν has no sign; a negative λ is not an answer but a wrong active set.